Harnessing the NLP for Diabetic Text Classification using Neural Network

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Abstract. Diabetes, being a chronic disease, has the capacity to engender a global healthcare catastrophe. Integrating machine learning technology with medical science enables a precise prognosis of an individual's susceptibility to diabetes. This study develops an efficient machine learning model for predicting diabetes, analyzing symptom associations, their influence on prediction, and identifying the optimal ML approach. Diabetes remains a leading cause of mortality globally. According to the International Diabetes Alliance (IDA), 382 million people currently have diabetes and this number is expected to double to 592 million by 2035.

As a traditional ML model, The model chosen for this task is XGBoost.Xie & Xu (2024). The proposed model is evaluated in the GitHub database: prediction performance at each level was analyzed using the Precision, Recall, accuracy and F1 metrics, and overall prediction efficiency was assessed using accuracy and its macro-average, weighted avg. The overall accuracy is obtained by this model 76 percent respectively. The experimental results shows that is more effective at predicting diabetes.

1 Introduction

1.1 Problem Statement

Diabetes occurs when the pancreas produces insufficient insulin or the body can't use it effectively, leading to high blood sugar. Uncontrolled diabetes can cause severe damage to nerves, blood vessels, and organs, increasing the risk of complications like heart disease, eye problems, and kidney failure. Early and accurate prediction is crucial for improving patient outcomes and reducing long-term risks.

1.2 Project Goals

This research proposes a neural network (NN) model to predict diabetes using clinical attributes.Wu (2023) The primary objective of this research is to design, implement, and evaluate predictive models using machine learning (ML) and deep learning (DL) techniques to improve the precision of diabetes diagnosis. The proposed research aligns with the global need for improved diagnostic tools to manage the increasing prevalence of diabetes. Using AI-based approaches, the study aims to help healthcare professionals make informed decisions, reduce complications, and improve patient outcomes.

2 Background

According to the World Health Organization (WHO), diabetes has seen a sharp rise in prevalence, with over 422 million individuals affected globally. The need for early diagnosis and intervention is critical to managing this growing epidemic. This section outlines the algorithms utilized in this study for diabetic text classification, emphasizing their theoretical underpinnings and practical implementations:

2.1 Support Vector Machine

Support Vector Machines (SVMs) excel in high-dimensional classification by handling linear and non-linear separability through kernels like Linear, Polynomial, RBF, and Sigmoid. Key hyperparameters (C, gamma, and degree) were optimized using grid search with cross-validation to improve generalization and minimize overfitting.Yang (2024)

$$\begin{aligned} \text{Linear: } K(w,b) &= w^T x + b \\ \text{Polynomial: } K(w,x) &= \left(\gamma w^T x + b\right)^N \\ \text{Gaussian RBF: } K(w,x) &= \exp\left(-\gamma \left\|x_i - x_j\right\|^n\right. \\ \text{Sigmoid}: K\left(x_i, x_j\right) &= \tanh\left(\alpha x_i^T x_j + b\right) \end{aligned} \tag{1}$$

Then finds out the process to separate the data based on the labels or outputs defined.*Support Vector Machine (SVM) Algorithm - GeeksforGeeks — geeksforgeeks.org* (n.d.)

2.2 Logistic Regression

Logistic Regression (LR) was used for binary classification, like predicting diabetes, by estimating probabilities with the sigmoid function. Two versions were tested: custom and Scikit-Learn's. Hyperparameters, including regularization strength (C) and penalty type (L1/L2), were optimized using grid search with cross-validation.

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Learning and validation curves assessed model performance and potential overfitting or underfitting.

$$p(X;b,w) = \frac{e^{w\cdot X+b}}{1+e^{w\cdot X+b}} = \frac{1}{1+e^{-w\cdot X+b}} \tag{2}$$

This function is known as the logistic function.*Logistic Regression in Machine Learning - GeeksforGeeks — geeksforgeeks.org* (n.d.)

2.3 Random Forest

Random Forest (RF) was used for diabetes prediction by combining decision trees to improve accuracy and reduce overfitting. Hyperparameters were optimized with Randomized Search Cross-Validation. Performance was evaluated with precision, recall, confusion matrix, and learning curves. SHAP was used for interpretability, highlighting feature impacts.

$$\hat{f}_{\mathrm{rf}}^{B}(x) = \frac{1}{B}\sum_{b=1}^{B} T\left(x;\Theta_b\right) \tag{3}$$

The random forest predictor formula.*math.mcgill.ca* (n.d.)

3 Methodology

3.1 Data Handling

3.1.1 Data Collection and Representation:

The dataset was sourced from GitHub, containing diagnostic attributes for predicting diabetes. The target variable, "Outcome," is binary, indicating whether a person has diabetes (1) or not (0).

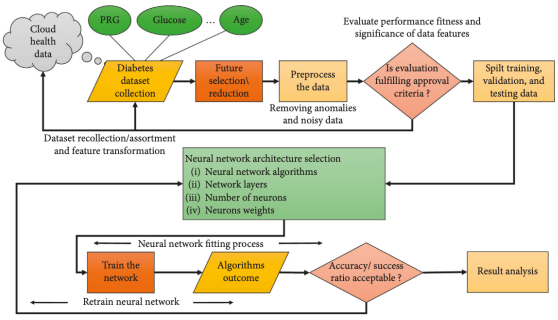

Figure 1.

3.2 Feature Selection/Reduction:

All provided features were considered, representing critical diagnostic factors such as glucose levels, BMI, and blood pressure. StandardScaler was applied to ensure uniform scaling, crucial for models sensitive to feature magnitudes, like Neural Networks and XGBoost.

$$tau = \frac{N_c - N_d}{n/2(n-1)}N \tag{4}$$

Here, tau represents the Kendall's correlation coefficient, Nc is the number of concordant pairs, Nd is the number of discordant pairs, and n is the total number of pairs.Qi et al. (2024)

3.2.1 Preprocessing:

The dataset underwent preprocessing, including splitting into training and test sets (80:20 ratio) and standardization of features to enhance model performance and stability.

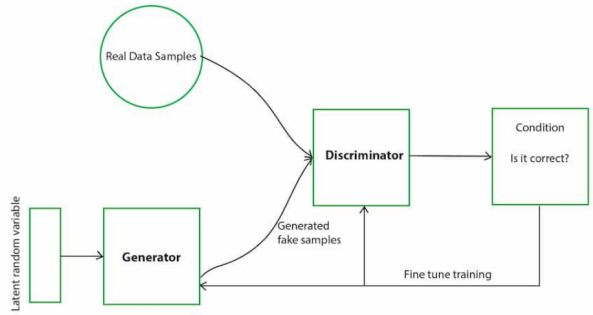

Figure 2.

3.3 Proposed Model : Neural Network

The neural networks consist of interconnected nodes or neurons that process and learn from data, enabling tasks such as pattern recognition and decision making in machine learning.

A neural network for email classification processes features like content, sender, and subject through hidden layers, learning patterns to identify spam. Using a binary activation function in the output layer, it predicts whether an email is spam (1) or not (0), refining its accuracy through backpropagation. (n.d.*a*)

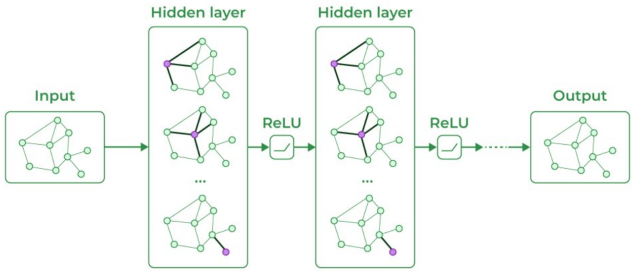

Figure 3.

The neural network algorithm was used independently for diabetes classification, alongside XGBoost, to compare their performance. The network's classification results were evaluated using metrics and visualizations, such as loss curves.

3.3.1 Relu Activation

$$f(x) = \max(0, x) \tag{5}$$

The ReLU function works great in most applications, and it is very widely used as a result.*Rectified Linear Units (ReLU) in Deep Learning — kaggle.com* (n.d.)

3.3.2 Sigmoid Activation

$$f(x) = 1/\left(1 + e^{\wedge}\left(-1^{\star}z\right)\right) \tag{6}$$

these just happen to be the most widely used. Also, the choice of the function is heavily used for Neural Networks.Babs (n.d.)

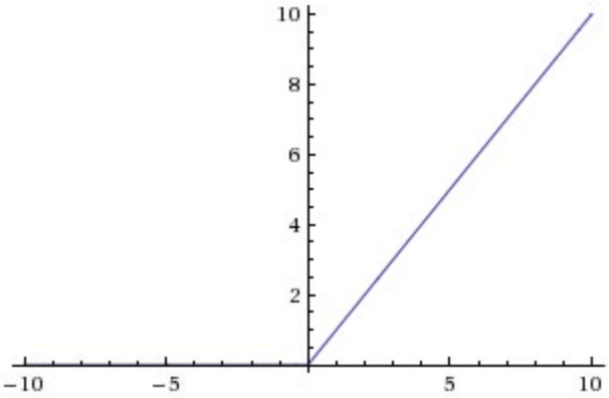

Figure 4.

3.4 Neural Network Fitting Process:

A neural network model was constructed using Keras Sequential API. The architecture included a 10-unit input layer with ReLU activation, a 3-unit hidden layer, and a sigmoid-activated output layer for binary classification. The model was compiled with the Adam optimizer and trained for 40 epochs with early stopping to prevent overfitting.

$$\text{Binary classification} : g(x) = \frac{1}{1 + e^{-\left(w_{K+1}x + b_{K+1}\right)}} \tag{7}$$

For a binary classification task, the output activation function is sigmoid. The value of the output will be between 0 and 1.Shi (n.d.)

3.5 Proposed System Model:

XGBoost was also employed to compare performance.Xie & Xu (2024) A grid search optimized its hyperparameters (learning rate, max depth, n estimators) using cross-validation. Learning curves and validation curves were plotted to evaluate training and generalization.

$$\text{Precise } = P = \frac{\mathrm{T}P}{\text{Predicted Positive}} = \frac{\mathrm{T}P}{\mathrm{TP} + \mathrm{FP}}$$
$$\text{Recall } = R = \frac{\mathrm{TP}}{\text{Actual Positive}} = \frac{TP}{TP + FN}$$
$$F1 = \frac{2}{\frac{l}{R} + \frac{1}{P}} = \frac{2 * P * R}{P + R} = \frac{2TP}{2TP + FP + FN}$$
$$\text{Accuracy } = A == \frac{\mathrm{T}P + TN}{\mathrm{T}P + FP + FR + TN}$$

The accuracy, precision, recall, F1 were used as assessment metrics in this study.Xie & Xu (2024)

3.6 Output Layer:

The final output layer in both models used a sigmoid activation to produce probabilities, interpreted as binary predictions for diabetes classification.

4 Experiments and results

4.1 Dataset info

The datasets consist of several medical predictors (independent) variables and one target (dependent) variable, Outcome. Independent variables include the number of pregnancies the patient has had, their BMI, insulin level, age, Diabetes pedigree function, Skin Thickness, Blood Pressure, Glucose, and Pregnancies.*Predict Diabetes From Medical Records — kaggle.com* (n.d.)

4.2 Designed experiments:

The experiments involved training and evaluating both XGBoost and Neural Network models on standardized data, with an 80/20 train-test split to compare performance.

4.3 Hyperparameter settings and Grid Search:

Grid search was used with 5-fold cross-validation to optimize learning rate, max depth, and n estimators for XGBoost, yielding the best configuration for model accuracy.

5 Discussion

In the various model metrics of this experiment, Neural network has better performance compared to other models, such as accuracy, recall, F1, and especially presicion. The model achieved an accuracy of 0.76, precision of 0.81, recall of 0.82, and F1 score of 0.81. Comparisons with Logistic Regression, SVM, and Random Forest focused on sampling, tree depth, handling of missing values, regularization, and computational efficiency. Hyperparameter optimization was conducted using grid search.

Neural network well performs on both positive and negative outcomes (n.d.*b*). it has flexibility in capturing complex patterns in the data. but the training time was longer compared to traditional models like logistic regression.

Logistic Regression result were comparable to Neural Networks. It showcased simplicity and interoperability *Logistic Regression in Machine Learning - GeeksforGeeks — geeksforgeeks.org* (n.d.). Linear separable data and got competitive performance. But limited capability in handling non-linear relationships compared to Neural Networks or Random Forest.

Support Vector Machine got lower accuracy to other models but got high Recall. The class distribution is imbalanced but performs well. But computational cost increases significantly with larger datasets *Support Vector Machine (SVM) Algorithm - GeeksforGeeks — geeksforgeeks.org* (n.d.).

Random Forest were slightly below Neural Networks. Through the features importance offers interpretability. Overfitting due to averaging is robustness. slightly lower Recall compared to Neural Networks.Wang (2024)

5.1 Difference of accuracy

Models	Accuracy Score	Precision	Recall	F1 Score
Neural Networks	0.76	0.81	0.82	0.81
Logistic Regression	0.75	0.81	0.80	0.81
Support Vector Machine	0.72	0.74	0.89	0.81
Random Forest	0.75	0.81	0.80	0.80

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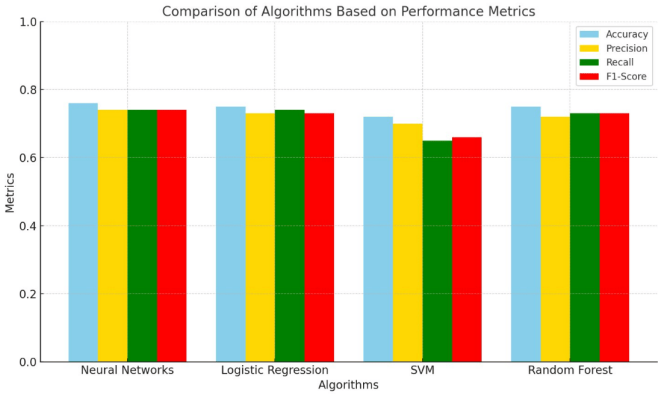

Figure 5.

6 Conclusion and future work

This study applied various machine learning techniques to classify data, with the Deep Neural Network outperforming traditional models. Data preprocessing, visualization, and hyperparameter optimization using GridSearchCV enhanced performance. The proposed model has potential future benefits for both general users and medical staff.

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